



Param Yuva II—fastest Indian supercomputer (Courtesy: <https://chevireddyhemachandradev.wordpress.com>)

German firm Lernstift has created a novel digital pen that will help people of all age groups to write legibly without making any mistake. Sensors built into the pen detect variations in the way letters are formed and also pick up errors in spellings. The pen vibrates immediately for misspelled words or letters formed illegibly.

The device functions in two modes: calligraphy and orthography. In calligraphy mode, the pen vibrates once if a letter is written wrongly or illegibly. In orthography mode, the pen vibrates once for a misspelled word and twice to point out grammatical errors in a sentence.

World's most energy-efficient supercomputer

Internationally-renowned Aurora supercomputer, built by Italy-based Eurotech Group, has set a new world record for data centre energy efficiency. The supercomputer, featuring NVIDIA Tesla GPU accelerators based on NVIDIA Kepler, reached 3150 megaflops per watt of sustained performance recently. This was 26 per cent higher than the efficiency of top supercomputers in the new Green500 list.

To set the new mark of efficiency, Aurora combined 128 high-performance, energy-efficient NVIDIA Tesla

K20 accelerators with the Eurotech Aurora Tigon supercomputer that uses Aurora hot-water cooling technology.

Informing about the benefits of the supercomputer's data centre energy efficiency, Sanzio Bassini, director of HPC department at the Cineca facility in Bologna, Italy, where the world record was set, said, "Advanced computer simulations that enable scientists to discover new phenomena and test hypotheses require massive amounts of performance, which can consume a lot of power."

Bassini added, "Equipped with the ultra-efficient Aurora system and NVIDIA GPU accelerators, Aurora will give European researchers the computing muscle to study all types of physical and biological systems while allowing us to keep data centre power consumption and costs in check."

India launches its fastest supercomputer

In a proud moment, India has launched its fastest supercomputer PARAM Yuva-II, which is a new 500-teraflop version of PARAM Yuva with peak computing power of 524 teraflops/s. This supercomputer is devised to solve large and complex computational problems, and it will help create a general-purpose research-oriented computational environment in the country.

Developed by C-DAC Pune, PARAM Yuva-II is expected to rank 62nd in the world and No. 1 in India. Interestingly, the increase in its peak compute power from 54 teraflops/s to 524 teraflops/s has been achieved without any significant increase in the electrical power consumption of the facility.

Wearable solar cell to recharge mobiles

Bengaluru-based firm Lumos is close to developing a wearable solar cell in the form of t-shirts or bags that would allow you to charge your mobiles, iPads, iPods and other such gadgets on the go.

During the research, the company successfully fitted this solar cell on a bag that also has a charging outlet and a small 1800mAh battery for storing the solar energy. The bag can also be charged with AC power supply and is made of water-resistant material. It is fitted with a wide range of connectors to charge different models of mobile phones.

India to have its own GPS soon

India will soon become independent of the USA-controlled global positioning system (GPS) as the country is all set to launch its first satellite-based navigation system 'GAGAN' this year. Announcing this recently, S. Ramakrishnan, director, Vikram Sarabhai Space Centre (VSSC), informed that India's first series of navigation satellites will aid in providing regional navigation service and ensure the country's entry into the elite club of nations like USA, Russia and China that have their own GPS.

However, before the service of 'GAGAN' (GPS Aided Geo Augmented Navigation) could be started, it is necessary to launch a number of satellites. The first satellite of this series—named 'Indian regional navigation satellite system' (IRNSS)—is likely to be launched by PSLV C-22 rocket in the second half of this year.

Ramakrishnan informed that once all the required satellites were launched, GAGAN would start providing navigation service most probably in 2014.

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